

6.8.44 INT_SOURCE Register (Offset = 2Bh) [Reset = 00h]

INT_SOURCE is shown in [Figure 6-61](#) and described in [Table 6-53](#).

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Figure 6-61. INT_SOURCE Register

7	6	5	4	3	2	1	0
INT_PB_IS_SET	INT_LDO_3_4_IS_SET	INT_LDO_1_2_IS_SET	INT_BUCK_3_IS_SET	INT_BUCK_1_2_IS_SET	INT_SYSTEM_IS_SET	INT_RV_IS_SET	INT_TIMEOUT_RV_SD_IS_SET
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h

Table 6-53. INT_SOURCE Register Field Descriptions

Bit	Field	Type	Reset	Description
7	INT_PB_IS_SET	R	0h	One or more sources of the INT present in register INT_PB 0h = No bits set in INT_PB 1h = One or more bits set in INT_PB
6	INT_LDO_3_4_IS_SET	R	0h	One or more sources of the INT present in register INT_LDO_3_4 0h = No bits set in INT_LDO_3_4 1h = One or more bits set in INT_LDO_3_4
5	INT_LDO_1_2_IS_SET	R	0h	One or more sources of the INT present in register INT_LDO_1_2 0h = No bits set in INT_LDO_1_2 1h = One or more bits set in INT_LDO_1_2
4	INT_BUCK_3_IS_SET	R	0h	One or more sources of the INT present in register INT_BUCK_3 0h = No bits set in INT_BUCK_3 1h = One or more bits set in INT_BUCK_3
3	INT_BUCK_1_2_IS_SET	R	0h	One or more sources of the INT present in register INT_BUCK_1_2 0h = No bits set in INT_BUCK_1_2 1h = One or more bits set in INT_BUCK_1_2
2	INT_SYSTEM_IS_SET	R	0h	One or more sources of the INT present in register INT_SYSTEM 0h = No bits set in INT_SYSTEM 1h = One or more bits set in INT_SYSTEM
1	INT_RV_IS_SET	R	0h	One or more sources of the INT present in register INT_RV 0h = No bits set in INT_RV 1h = One or more bits set in INT_RV
0	INT_TIMEOUT_RV_SD_IS_SET	R	0h	One or more sources of the INT present in register INT_TIMEOUT_RV_SD 0h = No bits set in INT_TIMEOUT_RV_SD 1h = One or more bits set in INT_TIMEOUT_RV_SD

6.8.45 INT_LDO_3_4 Register (Offset = 2Ch) [Reset = X]

INT_LDO_3_4 is shown in [Figure 6-62](#) and described in [Table 6-54](#).

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Figure 6-62. INT_LDO_3_4 Register

7	6	5	4	3	2	1	0
RESERVED	RESERVED	LDO4_UV	LDO4_OC	LDO4_SCG	LDO3_UV	LDO3_OC	LDO3_SCG
R-X	R-X	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h

Table 6-54. INT_LDO_3_4 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	RESERVED	R	X	Reserved
6	RESERVED	R	X	Reserved
5	LDO4_UV	R/W1C	0h	LDO4 Undervoltage Fault. Is automatically cleared upon a transition to INITIALIZE state, if corresponding *_UV_MASK bit in register INT_MASK_UV is '1' 0h = No Fault detected 1h = Fault detected
4	LDO4_OC	R/W1C	0h	LDO4 Overcurrent Fault. 0h = No Fault detected 1h = Fault detected
3	LDO4_SCG	R/W1C	0h	LDO4 Short Circuit to Ground Fault 0h = No Fault detected 1h = Fault detected
2	LDO3_UV	R/W1C	0h	LDO3 Undervoltage Fault. Is automatically cleared upon a transition to INITIALIZE state, if corresponding *_UV_MASK bit in register INT_MASK_UV is '1' 0h = No Fault detected 1h = Fault detected
1	LDO3_OC	R/W1C	0h	LDO3 Overcurrent Fault 0h = No Fault detected 1h = Fault detected
0	LDO3_SCG	R/W1C	0h	LDO3 Short Circuit to Ground Fault 0h = No Fault detected 1h = Fault detected

6.8.46 INT_LDO_1_2 Register (Offset = 2Dh) [Reset = X]

INT_LDO_1_2 is shown in [Figure 6-63](#) and described in [Table 6-55](#).

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Figure 6-63. INT_LDO_1_2 Register

7	6	5	4	3	2	1	0
RESERVED	RESERVED	LDO2_UV	LDO2_OC	LDO2_SCG	LDO1_UV	LDO1_OC	LDO1_SCG
R-X	R-X	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h

Table 6-55. INT_LDO_1_2 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	RESERVED	R	X	Reserved
6	RESERVED	R	X	Reserved
5	LDO2_UV	R/W1C	0h	LDO2 Undervoltage Fault. Is automatically cleared upon a transition to INITIALIZE state, if corresponding *_UV_MASK bit in register INT_MASK_UV is '1' 0h = No Fault detected 1h = Fault detected
4	LDO2_OC	R/W1C	0h	LDO2 Overcurrent Fault 0h = No Fault detected 1h = Fault detected
3	LDO2_SCG	R/W1C	0h	LDO2 Short Circuit to Ground Fault 0h = No Fault detected 1h = Fault detected
2	LDO1_UV	R/W1C	0h	LDO1 Undervoltage Fault. Is automatically cleared upon a transition to INITIALIZE state, if corresponding *_UV_MASK bit in register INT_MASK_UV is '1' 0h = No Fault detected 1h = Fault detected
1	LDO1_OC	R/W1C	0h	LDO1 Overcurrent Fault 0h = No Fault detected 1h = Fault detected
0	LDO1_SCG	R/W1C	0h	LDO1 Short Circuit to Ground Fault 0h = No Fault detected 1h = Fault detected

6.8.47 INT_BUCK_3 Register (Offset = 2Eh) [Reset = X]

INT_BUCK_3 is shown in [Figure 6-64](#) and described in [Table 6-56](#).

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Figure 6-64. INT_BUCK_3 Register

7	6	5	4	3	2	1	0
RESERVED	RESERVED	RESERVED	RESERVED	BUCK3_UV	BUCK3_NEG_OC	BUCK3_OC	BUCK3_SCG
R-X	R-X	R-X	R-X	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h

Table 6-56. INT_BUCK_3 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	RESERVED	R	X	Reserved
6	RESERVED	R	X	Reserved
5	RESERVED	R	X	Reserved
4	RESERVED	R	X	Reserved
3	BUCK3_UV	R/W1C	0h	BUCK3 Undervoltage Fault. Is automatically cleared upon a transition to INITIALIZE state, if corresponding *_UV_MASK bit in register INT_MASK_UV is '1' 0h = No Fault detected 1h = Fault detected
2	BUCK3_NEG_OC	R/W1C	0h	BUCK3 Negative Overcurrent Fault 0h = No Fault detected 1h = Fault detected
1	BUCK3_OC	R/W1C	0h	BUCK3 Positive Overcurrent Fault 0h = No Fault detected 1h = Fault detected
0	BUCK3_SCG	R/W1C	0h	BUCK3 Short Circuit to Ground Fault 0h = No Fault detected 1h = Fault detected

6.8.48 INT_BUCK_1_2 Register (Offset = 2Fh) [Reset = 00h]

INT_BUCK_1_2 is shown in [Figure 6-65](#) and described in [Table 6-57](#).

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Figure 6-65. INT_BUCK_1_2 Register

7	6	5	4	3	2	1	0
BUCK2_UV	BUCK2_NEG_OC	BUCK2_OC	BUCK2_SCG	BUCK1_UV	BUCK1_NEG_OC	BUCK1_OC	BUCK1_SCG
R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h

Table 6-57. INT_BUCK_1_2 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	BUCK2_UV	R/W1C	0h	BUCK2 Undervoltage Fault. Is automatically cleared upon a transition to INITIALIZE state, if corresponding *_UV_MASK bit in register INT_MASK_UV is '1' 0h = No Fault detected 1h = Fault detected
6	BUCK2_NEG_OC	R/W1C	0h	BUCK2 Negative Overcurrent Fault 0h = No Fault detected 1h = Fault detected
5	BUCK2_OC	R/W1C	0h	BUCK2 Positive Overcurrent Fault 0h = No Fault detected 1h = Fault detected
4	BUCK2_SCG	R/W1C	0h	BUCK2 Short Circuit to Ground Fault 0h = No Fault detected 1h = Fault detected
3	BUCK1_UV	R/W1C	0h	BUCK1 Undervoltage Fault. Is automatically cleared upon a transition to INITIALIZE state, if corresponding *_UV_MASK bit in register INT_MASK_UV is '1' 0h = No Fault detected 1h = Fault detected
2	BUCK1_NEG_OC	R/W1C	0h	BUCK1 Negative Overcurrent Fault 0h = No Fault detected 1h = Fault detected
1	BUCK1_OC	R/W1C	0h	BUCK1 Positive Overcurrent Fault 0h = No Fault detected 1h = Fault detected
0	BUCK1_SCG	R/W1C	0h	BUCK1 Short Circuit to Ground Fault 0h = No Fault detected 1h = Fault detected

6.8.49 INT_SYSTEM Register (Offset = 30h) [Reset = 00h]

INT_SYSTEM is shown in [Figure 6-66](#) and described in [Table 6-58](#).

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Figure 6-66. INT_SYSTEM Register

7	6	5	4	3	2	1	0
SENSOR_0_H OT	SENSOR_1_H OT	SENSOR_2_H OT	SENSOR_3_H OT	SENSOR_0_W ARM	SENSOR_1_W ARM	SENSOR_2_W ARM	SENSOR_3_W ARM
R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h

Table 6-58. INT_SYSTEM Register Field Descriptions

Bit	Field	Type	Reset	Description
7	SENSOR_0_HOT	R/W1C	0h	TSD Hot detection for sensor 0 0h = No Fault detected 1h = Fault detected
6	SENSOR_1_HOT	R/W1C	0h	TSD Hot detection for sensor 1 0h = No Fault detected 1h = Fault detected
5	SENSOR_2_HOT	R/W1C	0h	TSD Hot detection for sensor 2 0h = No Fault detected 1h = Fault detected
4	SENSOR_3_HOT	R/W1C	0h	TSD Hot detection for sensor 3 0h = No Fault detected 1h = Fault detected
3	SENSOR_0_WARM	R/W1C	0h	TSD Warm detection for sensor 0. Is automatically cleared upon a transition to INITIALIZE state, if corresponding *_WARM_MASK bit in register MASK_CONFIG is '1' 0h = No Fault detected 1h = Fault detected
2	SENSOR_1_WARM	R/W1C	0h	TSD Warm detection for sensor 1. Is automatically cleared upon a transition to INITIALIZE state, if corresponding *_WARM_MASK bit in register MASK_CONFIG is '1' 0h = No Fault detected 1h = Fault detected
1	SENSOR_2_WARM	R/W1C	0h	TSD Warm detection for sensor 2. Is automatically cleared upon a transition to INITIALIZE state, if corresponding *_WARM_MASK bit in register MASK_CONFIG is '1' 0h = No Fault detected 1h = Fault detected
0	SENSOR_3_WARM	R/W1C	0h	TSD Warm detection for sensor 3. Is automatically cleared upon a transition to INITIALIZE state, if corresponding *_WARM_MASK bit in register MASK_CONFIG is '1' 0h = No Fault detected 1h = Fault detected

6.8.50 INT_RV Register (Offset = 31h) [Reset = X]

INT_RV is shown in [Figure 6-67](#) and described in [Table 6-59](#).

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Figure 6-67. INT_RV Register

7	6	5	4	3	2	1	0
RESERVED	LDO4_RV	LDO3_RV	LDO2_RV	LDO1_RV	BUCK3_RV	BUCK2_RV	BUCK1_RV
R-X	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h

Table 6-59. INT_RV Register Field Descriptions

Bit	Field	Type	Reset	Description
7	RESERVED	R	X	Reserved
6	LDO4_RV	R/W1C	0h	RV event detected on LDO4 rail during rail-turn-on, or after 4-5 ms during discharge checks prior to entering power sequence to ACTIVE state 0h = No RV detected 1h = RV detected
5	LDO3_RV	R/W1C	0h	RV event detected on LDO3 rail during rail-turn-on, or after 4-5 ms during discharge checks prior to entering power sequence to ACTIVE state 0h = No RV detected 1h = RV detected
4	LDO2_RV	R/W1C	0h	RV event detected on LDO2 rail during rail-turn-on, or after 4-5 ms during discharge checks prior to entering power sequence to ACTIVE state 0h = No RV detected 1h = RV detected
3	LDO1_RV	R/W1C	0h	RV event detected on LDO1 rail during rail-turn-on, or after 4-5 ms during discharge checks prior to entering power sequence to ACTIVE state 0h = No RV detected 1h = RV detected
2	BUCK3_RV	R/W1C	0h	RV event detected on BUCK3 rail during rail-turn-on, or after 4-5 ms during discharge checks prior to entering power sequence to ACTIVE state 0h = No RV detected 1h = RV detected
1	BUCK2_RV	R/W1C	0h	RV event detected on BUCK2 rail during rail-turn-on, or after 4-5 ms during discharge checks prior to entering power sequence to ACTIVE state 0h = No RV detected 1h = RV detected
0	BUCK1_RV	R/W1C	0h	RV event detected on BUCK1 rail during rail-turn-on, or after 4-5 ms during discharge checks prior to entering power sequence to ACTIVE state 0h = No RV detected 1h = RV detected

6.8.51 INT_TIMEOUT_RV_SD Register (Offset = 32h) [Reset = 00h]

INT_TIMEOUT_RV_SD is shown in [Figure 6-68](#) and described in [Table 6-60](#).

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Figure 6-68. INT_TIMEOUT_RV_SD Register

7	6	5	4	3	2	1	0
TIMEOUT	LDO4_RV_SD	LDO3_RV_SD	LDO2_RV_SD	LDO1_RV_SD	BUCK3_RV_SD	BUCK2_RV_SD	BUCK1_RV_SD
R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h	R/W1C-0h

Table 6-60. INT_TIMEOUT_RV_SD Register Field Descriptions

Bit	Field	Type	Reset	Description
7	TIMEOUT	R/W1C	0h	Is set if ShutDown occurred due to a TimeOut while: 1. Transitioning to ACTIVE state, and one or more rails did not rise past the UV level at the end of the assigned slot (and UV on this rail is configured as a SD fault). Which rail(s) is/are indicated by the *_UV bits in the INT_* registers. 2. Transitioning to STANDBY state, and one or more rails did not fall below the SCG level at the end of the assigned slot and discharge is enabled for that rail (which rail(s) is/are indicated by the corresponding RV_SD bit(s) in this register). 0h = No SD due to TimeOut occurred 1h = SD due to TimeOut occurred
6	LDO4_RV_SD	R/W1C	0h	RV on LDO4 rail caused a shutdown during: 1. A transition to STANDBY state, this rail did not discharge at the end of the assigned slot and discharge is enabled for this rail 2. A transition to STANDBY state, RV was observed on this rail during the transition after this rail was disabled and discharge was enabled 3. A transition to ACTIVE state, RV was observed on this rail during the transition when this rail was OFF (rails are expected to be discharged before commencing the sequence to ACTIVE) 4. This rail did not discharge and therefore caused a Timeout-SD while attempting to discharge all rails at the start of a transition from STANDBY to ACTIVE (TIMEOUT bit gets also set in this case) 0h = No SD due to RV/DISCHARGE_TIMEOUT on LDO4 occurred 1h = SD due to RV/DISCHARGE_TIMEOUT on LDO4 occurred
5	LDO3_RV_SD	R/W1C	0h	RV on LDO4 rail caused a shutdown during: 1. A transition to STANDBY state, this rail did not discharge at the end of the assigned slot and discharge is enabled for this rail 2. A transition to STANDBY state, RV was observed on this rail during the transition after this rail was disabled and discharge was enabled 3. A transition to ACTIVE state, RV was observed on this rail during the transition when this rail was OFF (rails are expected to be discharged before commencing the sequence to ACTIVE) 4. This rail did not discharge and therefore caused a Timeout-SD while attempting to discharge all rails at the start of a transition from STANDBY to ACTIVE (TIMEOUT bit gets also set in this case) 0h = No SD due to RV/DISCHARGE_TIMEOUT on LDO3 occurred 1h = SD due to RV/DISCHARGE_TIMEOUT on LDO3 occurred
4	LDO2_RV_SD	R/W1C	0h	RV on LDO4 rail caused a shutdown during: 1. A transition to STANDBY state, this rail did not discharge at the end of the assigned slot and discharge is enabled for this rail 2. A transition to STANDBY state, RV was observed on this rail during the transition after this rail was disabled and discharge was enabled 3. A transition to ACTIVE state, RV was observed on this rail during the transition when this rail was OFF (rails are expected to be discharged before commencing the sequence to ACTIVE) 4. This rail did not discharge and therefore caused a Timeout-SD while attempting to discharge all rails at the start of a transition from STANDBY to ACTIVE (TIMEOUT bit gets also set in this case) 0h = No SD due to RV/DISCHARGE_TIMEOUT on LDO2 occurred 1h = SD due to RV/DISCHARGE_TIMEOUT on LDO2 occurred

Table 6-60. INT_TIMEOUT_RV_SD Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
3	LDO1_RV_SD	R/W1C	0h	RV on LDO4 rail caused a shutdown during: 1. A transition to STANDBY state, this rail did not discharge at the end of the assigned slot and discharge is enabled for this rail 2. A transition to STANDBY state, RV was observed on this rail during the transition after this rail was disabled and discharge was enabled 3. A transition to ACTIVE state, RV was observed on this rail during the transition when this rail was OFF (rails are expected to be discharged before commencing the sequence to ACTIVE) 4. This rail did not discharge and therefore caused a Timeout-SD while attempting to discharge all rails at the start of a transition from STANDBY to ACTIVE (TIMEOUT bit gets also set in this case) 0h = No SD due to RV/DISCHARGE_TIMEOUT on LDO1 occurred 1h = SD due to RV/DISCHARGE_TIMEOUT on LDO1 occurred
2	BUCK3_RV_SD	R/W1C	0h	RV on LDO4 rail caused a shutdown during: 1. A transition to STANDBY state, this rail did not discharge at the end of the assigned slot and discharge is enabled for this rail 2. A transition to STANDBY state, RV was observed on this rail during the transition after this rail was disabled and discharge was enabled 3. A transition to ACTIVE state, RV was observed on this rail during the transition when this rail was OFF (rails are expected to be discharged before commencing the sequence to ACTIVE) 4. This rail did not discharge and therefore caused a Timeout-SD while attempting to discharge all rails at the start of a transition from STANDBY to ACTIVE (TIMEOUT bit gets also set in this case) 0h = No SD due to RV/DISCHARGE_TIMEOUT on BUCK3 occurred 1h = SD due to RV/DISCHARGE_TIMEOUT on BUCK3 occurred
1	BUCK2_RV_SD	R/W1C	0h	RV on LDO4 rail caused a shutdown during: 1. A transition to STANDBY state, this rail did not discharge at the end of the assigned slot and discharge is enabled for this rail 2. A transition to STANDBY state, RV was observed on this rail during the transition after this rail was disabled and discharge was enabled 3. A transition to ACTIVE state, RV was observed on this rail during the transition when this rail was OFF (rails are expected to be discharged before commencing the sequence to ACTIVE) 4. This rail did not discharge and therefore caused a Timeout-SD while attempting to discharge all rails at the start of a transition from STANDBY to ACTIVE (TIMEOUT bit gets also set in this case) 0h = No SD due to RV/DISCHARGE_TIMEOUT on BUCK2 occurred 1h = SD due to RV/DISCHARGE_TIMEOUT on BUCK2 occurred
0	BUCK1_RV_SD	R/W1C	0h	RV on LDO4 rail caused a shutdown during: 1. A transition to STANDBY state, this rail did not discharge at the end of the assigned slot and discharge is enabled for this rail 2. A transition to STANDBY state, RV was observed on this rail during the transition after this rail was disabled and discharge was enabled 3. A transition to ACTIVE state, RV was observed on this rail during the transition when this rail was OFF (rails are expected to be discharged before commencing the sequence to ACTIVE) 4. This rail did not discharge and therefore caused a Timeout-SD while attempting to discharge all rails at the start of a transition from STANDBY to ACTIVE (TIMEOUT bit gets also set in this case) 0h = No SD due to RV/DISCHARGE_TIMEOUT on BUCK1 occurred 1h = SD due to RV/DISCHARGE_TIMEOUT on BUCK1 occurred